

Chaemin Lim

PH.D. CANDIDATE IN COMPUTER SCIENCE · DATABASE SYSTEMS RESEARCHER

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Research Interests

My research broadly spans systems software, with a primary focus on database systems. I design hardware-aware execution and data-movement mechanisms for heterogeneous platforms, ranging from GPU-, PIM-, and DPU-accelerated servers to resource-constrained and trusted edge devices. My goal is to translate emerging hardware capabilities into practical, efficient, and secure end-to-end systems.

- **GPU-accelerated databases:** analytical query engines, relational OLAP, data movement, and out-of-memory execution.
- **Near-data processing systems:** join processing, skew handling, and collective communication on commodity PIM hardware, with on-going work on DPU-offloaded database processing.
- **Systems software for secure and resource-constrained edge computing:** trusted execution environments, efficient on-device inference, and attack-resistant IoT platforms.

Publications

1. **MEPID-Join: Scaling Processing-in-DIMM Joins Using Adaptive Key-Level Replication**
Suhyun Lee, Hanna Cha, **Chaemin Lim**, Hanjun Kim, and Youngsok Kim.
53rd International Conference on Very Large Data Bases (VLDB), Aug. 2027 (to appear).
Keywords: PIM; Database Systems; Join Processing
2. **GJP: Fast and Efficient Join-Project Query Processing on GPUs Using Density-Aware Matrix Multiplication**
Sungchul Lee, Suhyun Lee, Hanna Cha, **Chaemin Lim**, and Youngsok Kim.
2027 ACM International Conference on Management of Data (SIGMOD), June 2027 (to appear).
Keywords: GPU; Database Systems; Join-Project Processing; Matrix Multiplication
3. **FaScalSQL: A Fast and Scalable GPU-Accelerated SQL Query Engine for Out-of-Memory Tables**
Chaemin Lim, Suhyun Lee, Jinwoo Choi, Kwanghyun Park, Jinho Lee, Joonsung Kim, and Youngsok Kim.
42nd IEEE International Conference on Data Engineering (ICDE), May 2026. [Github]
Keywords: GPU; Database Systems; Out-of-Memory Query Processing
4. **DMO-DB: Mitigating the Data Movement Bottlenecks of GPU-Accelerated Relational OLAP**
Chaemin Lim, Suhyun Lee, Jinwoo Choi, Joonsung Kim, Jinho Lee, and Youngsok Kim.
34th International Conference on Parallel Architectures and Compilation Techniques (PACT), Nov. 2025.
Keywords: GPU; Database Systems; Data Movement; Relational OLAP
5. **SPID-Join: A Skew-resistant Processing-in-DIMM Join Algorithm Exploiting the Bank- and Rank-level Parallelisms of DIMMs**
Suhyun Lee, **Chaemin Lim**, Jinwoo Choi, Heelim Choi, Chan Lee, Yongjun Park, Kwanghyun Park, Hanjun Kim, and Youngsok Kim.
2025 ACM International Conference on Management of Data (SIGMOD), June 2025. [Github]
Keywords: PIM; Database Systems; Join Processing; Data Skew
6. **PID-Comm: A Fast and Flexible Collective Communication Framework for Commodity Processing-in-DIMMs**
Si Ung Noh¹, Junguk Hong¹, **Chaemin Lim**, Seongyeon Park, Jeehyun Kim, Hanjun Kim, Youngsok Kim, and Jinho Lee. ¹First Co-Author.
51st IEEE/ACM International Symposium on Computer Architecture (ISCA), June–July 2024. [Github]
Keywords: PIM; Collective Communication; Distributed Systems
7. **Design and Analysis of a Processing-in-DIMM Join Algorithm: A Case Study with UPMEM DIMMs**
Chaemin Lim, Suhyun Lee, Jinwoo Choi, Jounghoo Lee, Seongyeon Park, Hanjun Kim, Jinho Lee, and Youngsok Kim.
2023 ACM SIGMOD International Conference on Management of Data (SIGMOD), June 2023. [Github]
Keywords: PIM; Database Systems; Join Processing
8. **GuardiaNN: Fast and Secure On-Device Inference in TrustZone Using Embedded SRAM and Cryptographic Hardware**
Jinwoo Choi¹, Jaeyeon Kim¹, **Chaemin Lim**¹, Suhyun Lee, Jinho Lee, Dokyung Song, and Youngsok Kim. ¹First Co-Author.
23rd ACM/IFIP International Middleware Conference (Middleware), Nov. 2022.
Keywords: Trusted Execution Environments; On-Device Inference; Security

Education

Yonsei University

PH.D. CANDIDATE IN COMPUTER SCIENCE

- Advised by Prof. Youngsok Kim, High Performance Computing Platform (HPCP) Lab.

Seoul, Korea

Sep. 2023 – Present

Yonsei University

M.S. IN ARTIFICIAL INTELLIGENCE

- Advised by Prof. Youngsok Kim, High Performance Computing Platform (HPCP) Lab.

Seoul, Korea

Mar. 2021 – Aug. 2023

Hankuk University of Foreign Studies

B.A. IN JOURNALISM; B.ENG. IN COMPUTER ENGINEERING (DOUBLE MAJOR)

Seoul, Korea

Mar. 2017 – Feb. 2021

Patents

1. Skew Resistance Processing in a DIMM Device and Processing Method

Youngsok Kim, Suhyun Lee, Chaemin Lim, Jinwoo Choi, and Hanjun Kim.
US Patent Application 18/933,770, 2026.

2. Electronic Device and Method with Processing in Memory Operation

Bongjun Kim, Youngsok Kim, Sungchul Lee, Seil Lee, Suhyun Lee, Jounghoo Lee, Chaemin Lim, Hanna Cha, Jinwoo Choi, Yeonan Ha, and Hanwoong Jung.
US Patent Application 19/060,889, 2025.

3. PIM-Based Computing Device for Accelerating Join Operation

Youngsok Kim, Chaemin Lim, Hanjun Kim, and Jinho Lee.
US Patent 12,386,778, 2025.

4. Artificial Intelligence Device Based on Trust Environment

Youngsok Kim, Jinwoo Choi, Chaemin Lim, Suhyun Lee, Dokyung Song, and Jinho Lee.
US Patent Application 18/229,765, 2024.

Invited Talks & Conference Presentations

• FaScalSQL: A Fast and Scalable GPU-Accelerated SQL Query Engine for Out-of-Memory Tables

Conference Presenter, IEEE ICDE

May 2026

• DMO-DB: Mitigating the Data Movement Bottlenecks of GPU-Accelerated Relational OLAP

Conference Presenter, IEEE/ACM PACT

Nov. 2025

• A Fast Processing-in-DIMM Join Algorithm Exploiting UPMEM DIMMs

Invited Speaker, ABUMPIMP, EUROPAR

Aug. 2023

• Design and Analysis of a Processing-in-DIMM Join Algorithm: A Case Study with UPMEM DIMMs

Conference Presenter, ACM SIGMOD

June 2023

Awards

Samsung Humantech Paper Awards

Samsung Electronics

ENCOURAGEMENT PRIZE

2024

- Awarded for "PID-Comm: Fast and Flexible Collective Communication Framework for Commodity Processing-in-DIMM Devices."

Employment

Mangoboost

Seoul, Korea

SYSTEM ARCHITECT INTERN

Aug. 2025 – Jul. 2026

- Worked primarily on accelerating prefill-decode (PD) disaggregated LLM serving on AMD GPU clusters, focusing on RDMA-based KV-cache transfer, multi-node serving, and ROCm inference paths.
- Improved the MoRIIO serving path across asynchronous KV-transfer completion and cleanup, send draining, READ handoff routing, and multi-node TP=16 prefill-decode execution.
- Made selected upstream contributions to vLLM: AMD AITER MLA/MTP and FP8 correctness [#45227], shared-memory broadcast reliability [#45224], and MoRIIO proxy health monitoring [#45222].

Funded Research Projects

Development of New Concept PIM Semiconductor Leading Technology

IITP

2022 – Present

- Developed database and communication systems on commodity UPMEM DIMMs, including PID-Join, SPID-Join, and PID-Comm.
- Released reproducibility artifacts for join processing and collective communication.

STACK: Smart, Attack-Resistant Internet of Things Networks

KIAT

2021

- Worked with trusted execution environment software for attack-resistant IoT systems.
- Reported an STM32MP1 clock-rate computation bug in OP-TEE OS; the merged fix includes a **Reported-by** acknowledgment. [OP-TEE #4904]

Teaching Experience

Teaching Assistant, Logic Circuit Design (CSI2111), Yonsei University

Fall 2022

Teaching Assistant, Open Source Projects (CSI2115), Yonsei University

Fall 2021

Teaching Assistant, Information Security (CSI4109), Yonsei University

Spring 2021

Technical Skills

Programming: C, C++, Python, Scala, Rust, SQL